UNIVERSITY OF INFORMATION TECHNOLOGY AND SCIENCES

Lab Report

Course Title: Internet of Things Lab
Course Code: CSE 402

Experiment No:02
Experiment Name:Pushbutton digital input with Arduino.

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Objective:

1. To learn how pushbuttons function control simple electronic systems and understand real-world use cases of pushbuttons in embedded systems.
2. To construct a basic circuit using a pushbutton and resistor to ensure proper operation and prevent floating input states.
3. To write an Arduino sketch to read the pushbutton state and respond accordingly.
4. To explore the issue of switch bouncing and how to implement software/hardware debouncing techniques.

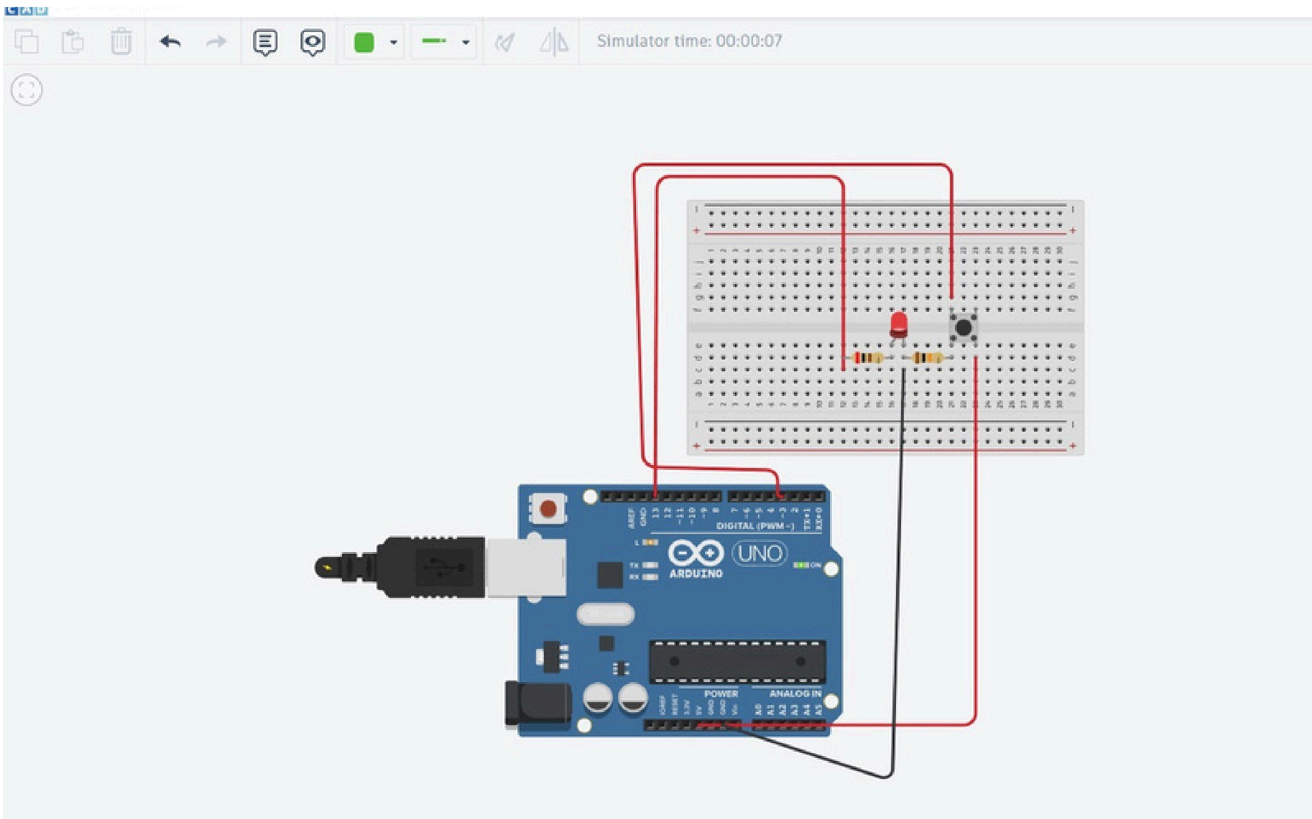
Introduction:

Pushbuttons are widely used as digital inputs in microcontroller-based systems. In this lab, a pushbutton is interfaced with an Arduino to detect button presses and control an output, such as an LED. The experiment covers essential concepts like pull-up/pull-down resistors and debouncing to ensure accurate signal detection. Understanding these principles is crucial for designing reliable user interfaces in embedded systems.

Apparatus Required:

Name	Quantity	Component
U2	1	Arduino Uno R3
S1	1	Pushbutton
R1	1	10 kΩ Resistor
D1	1	Red LED
R2	1	200 Ω Resistor

Circuit Diagram:



Working Procedure:

In this experiment, a pushbutton was connected to an Arduino as a digital input. The circuit was assembled by wiring one terminal of the pushbutton to a digital input pin and the other to the ground, with a pull-up resistor ensuring a stable HIGH signal when the button was not pressed. The Arduino was programmed to read the button state using the digitalRead() function and trigger an action, such as turning an LED on or off. After uploading the code, the button was tested for responsiveness, and any issues like bouncing were observed. To improve reliability, a debouncing technique was implemented using a short delay or state change detection in the code. The final setup ensured accurate and stable input detection, demonstrating the importance of pull-up resistors and debouncing in digital input systems.

Code:

```
const int buttonPin = 3;

const int ledPin = 13;

int buttonStatus = 0;

void setup()

{

   pinMode(buttonPin, INPUT);
   pinMode(ledPin, OUTPUT);

 }
 void loop()

 {    buttonStatus=digitalRead(buttonPin);
      if(buttonStatus == HIGH){
      digitalWrite(ledPin,HIGH);

}

else

{

digitalWrite(ledPin,LOW);

}
}
```

Conclusion:

This experiment successfully demonstrated how to interface a pushbutton with an Arduino as a digital input. By using a pull-up resistor, we ensured a stable signal and avoided floating states. The implementation of debouncing techniques improved the accuracy of button press detection, preventing unintended multiple triggers. Through this experiment, we gained a better understanding of digital input handling, which is essential for designing reliable user interfaces in embedded systems.